

Capstone 3: Enterprise RAG Assistant for Technical Knowledge

Siemens | Day 1 – RAG for Developer & Enterprise Knowledge Workflows | Estimated build time: 1–2 hours

Problem Statement

A Siemens engineering team needs fast, grounded answers from a controlled collection of technical standards, API specifications, architecture guidance, troubleshooting procedures, and internal engineering documentation. A generic LLM response is not sufficient because answers must be traceable to approved source material and should avoid exposing information outside the user's intended knowledge boundary.

Approach

- Assemble a small synthetic enterprise knowledge base representing approved technical documents.
- Define document metadata such as document type, product/domain, version, owner, and access classification.
- Chunk documents appropriately and create a retrieval workflow that selects relevant passages rather than sending entire documents to the model.
- Add metadata filtering and basic access-control boundaries.
- Generate answers that cite the retrieved source documents and explicitly identify when evidence is insufficient.
- Test the assistant against representative technical questions, including questions requiring multiple sources and questions with no supported answer.

Deliverable

A working RAG prototype, indexed knowledge set, retrieval strategy, sample queries, grounded responses with source references, evaluation results, and a short security/data-leakage assessment.

Enterprise Constraints

- Use synthetic or approved training data only; do not place confidential Siemens information into the exercise environment.
- Keep AI-generated changes bounded, reviewable, and reproducible.
- Use human approval for consequential actions and deterministic checks wherever possible.

- Document assumptions, validation steps, and known limitations.

Suggested Acceptance Criteria

- The participant can demonstrate the complete workflow end to end.
- The output is traceable to the defined inputs and acceptance criteria.
- Failure, ambiguity, or insufficient evidence is surfaced rather than silently invented.
- The solution demonstrates a measurable improvement in engineering productivity, quality, consistency, or control.